Classification of Sets by Cardinality

Today's Agenda

- Cardinality of Sets
- Cardinality of Natural Numbers
- Cardinality of Whole Numbers
- Cardinality of Even Numbers
- Cardinality of Odd Numbers
- Cardinality of Integers
- Cardinality of Rational Numbers
- Cardinality of Real Numbers

Why study set sizes

Most of the mathematical constructs are defined in terms of sets.

Previously we studied how sets can constructed and de-structured in various ways. It includes union, intersection, subsets etc.

We will now focus more on the count of elements.

Classification of Sets by Sizes - Preview

- Set
  - Finite Set
  - Infinite Sets
    - Countably Infinite Sets
    - Uncountably Infinite Sets

Equivalent Sets

Equivalent Sets - Two finite sets A and B are equivalent if their element count is same. i.e. n(A) = n(B)

Example: {5, 10, 15, 20, 25), & {a, e, i, o, u} are equivalent.

Cardinality

The **number of elements** in a set is called **cardinality** of that set.
It is denoted by **|A|** for the *number of elements* in set A.

$$S = \{ 2, 3, 5, 7, 11, 16 \} \quad \Rightarrow \quad |S| = 6$$

The use of term **cardinality** instead of ***size*** is

- Being precise over idea of *count of the elements*
- Other similar terms like *measure, length, boundary* etc.

Rapid fire

What is the cardinality of set

A $= \{ x \mid x \in \mathbb{N}, 3 < x \leq 10 \}$?

Rapid fire

What is the cardinality of set

A = $\{ x \mid x \in \mathbb{N} , 3 < x \leq 10 \}$**?**

Ans: 7

Rapid fire

What is the cardinality of set

E = { a, b, c, d, …., x, y, z **}?**

Ans:

Rapid fire

What is the cardinality of set

E = { a, b, c, d, …., x, y, z **}?**

Ans: 26

Rapid fire

What is the cardinality of $\emptyset$

Ans:

Rapid fire

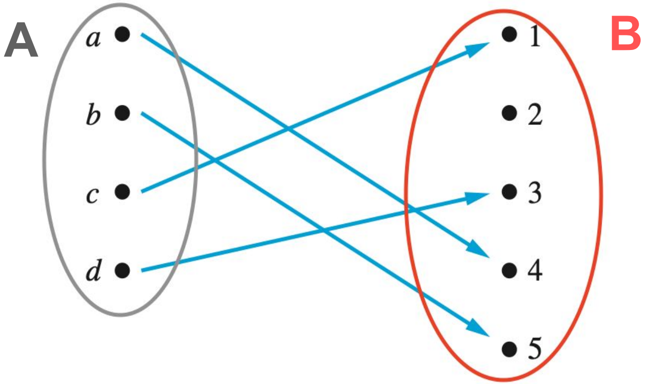
What is the cardinality of ∅

Ans: 0

Cardinality - Comparative

- If there exists an injective function from A to B,
  We say the cardinality of A is less than or same as B.

$$|A| \leq |B|$$



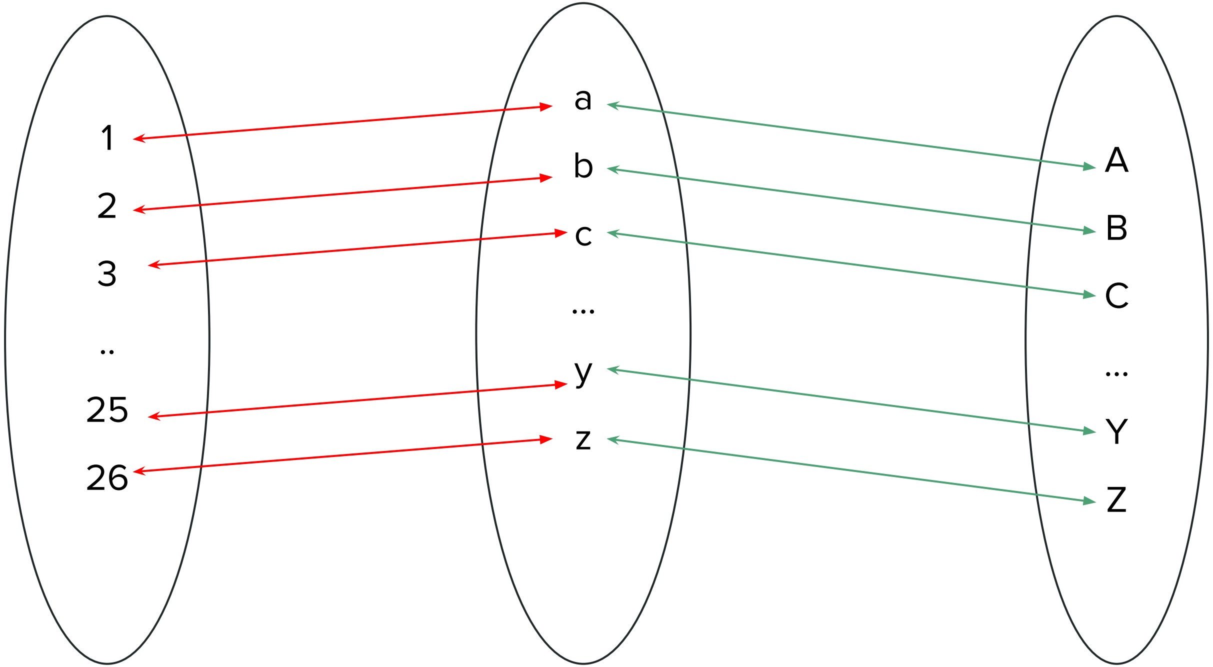
Cardinality - Equivalent

- If our injective function is also surjective, it becomes bijective, then A will have same number of elements as B, meaning same cardinality.

- The sets *A* and *B* have the **same *cardinality*** if and only if there is a **bijective function** (one-to-one correspondence) from *A* to *B*.

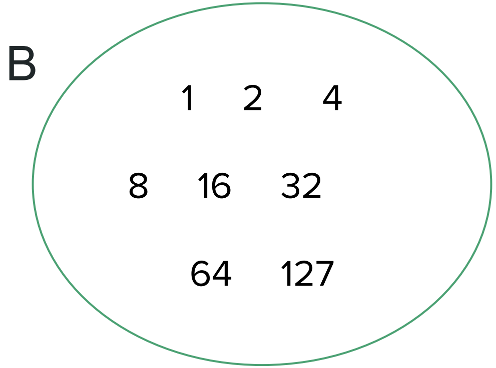
- When *A* and *B* have the same cardinality, we write $|A| = |B|$.

Cardinality - Equivalence Relation



Cardinality of Finite Sets

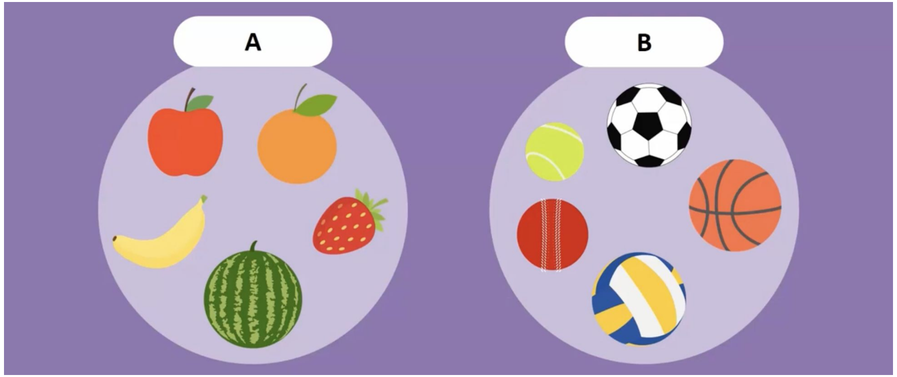
- A set with *no elements* is called a **null set**. i.e. ∅ . It's cardinality is 0.
- A **finite set** has a cardinality that can be represented by a nonnegative integer.



$|B| = 7$

Finite Vs Infinite Sets

Equivalent Sets - Two finite sets A and B are equivalent if their element count is same. i.e. $|A| = |B|$



Example:

$P = \{5, 10, 15, 20, 25\}$ and

$Q = \{a, e, i, o, u\}$

are equivalent.

We can easily compare two **finite sets** based on their .

But can we do the same for **infinite sets**?

Splitting the Naturals in two parts

$\mathbb{N}$
$\{ 1, 2, 3, 4, 5, \ldots \}$

O (odd)
$\{ 1, 3, 5, 7, \ldots \}$

E (even)
$\{ 2, 4, 6, 8, \ldots \}$

Try Answering

1. $|\mathbb{N}| = |E|$??
2. $|\mathbb{N}| = |O| + |E|$??
3. $|E| = |O|$??

Cardinality of Infinite Set

- An Infinite set is a set that is **not finite**.
- An infinite set is one that can be put into bijection with one of its proper subsets.

- The above statement feels counter-intuitive, but it can be shown with the help of famous Hilbert Hotel.

Hilbert Hotel - A short Visit

Room No	1	2	3	4	5	....
Day 1	A1	A2	A3	A4	A5	....

Hilbert Hotel - A short Visit

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Room No	1	2	3	4	5	….
Day 1	A1	A2	A3	A4	A5	….

Day 2	B1	B2

Hilbert Hotel - A short Visit

Room No	1	2	3	4	5	....
Day 1	A1	A2	A3	A4	A5	....

Day 2			A1	A2	A3	....

Hilbert Hotel - A short Visit

Room No	1	2	3	4	5	....
Day 1	A1	A2	A3	A4	A5	....

Day 2	B1	B2	A1	A2	A3	....

Hilbert Hotel - A short Visit

Room No	1	2	3	4	5	....
Day 1	A1	A2	A3	A4	A5	....

Day 2	B1	B2	A1	A2	A3	....

Hilbert Hotel - A short Visit

- Here set A is clearly a proper subset of set B

A	A1	A2	A3	A4	A5	….
B	B1	B2	A1	A2	A3	….

- Nonetheless, there is a bijection between A and B.
- This is the property of infinite set.

- Finite sets cannot have bijection if A is proper subset of B.

Cardinality of Natural Numbers

- We know that the size of $\mathbb{N}$ is **infinite**.
- Cardinality is typically discussed **between two sets**, by checking for a bijection.
- Therefore, when discussing the cardinality of **infinite sets**, we try to find a bijection between $\mathbb{N}$ and the set in question.

Now, let’s **revisit the odd-even split** of $\mathbb{N}$ to explore these ideas further.

Cardinality - Natural Vs Even Numbers

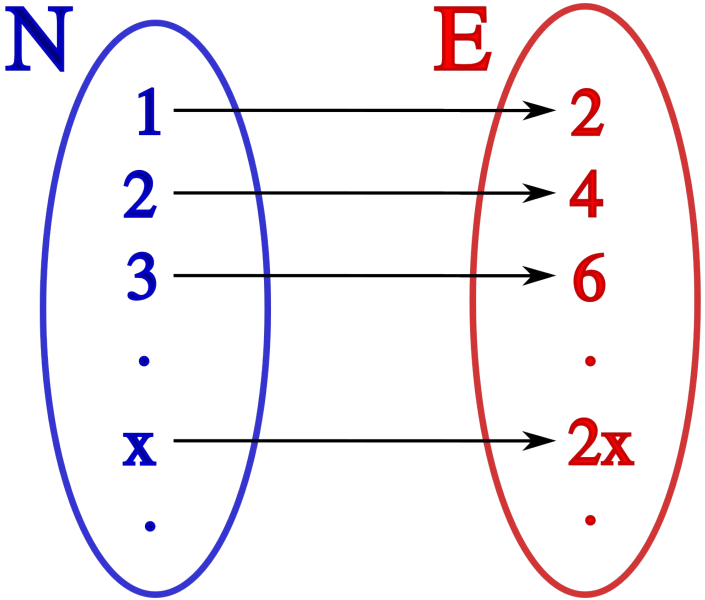
Lets try bijection from Natural numbers to Even numbers.

We can see that, with function **f(x) = 2x,**

For every natural number x (in set N),

there exists an even number 2x (in set E).

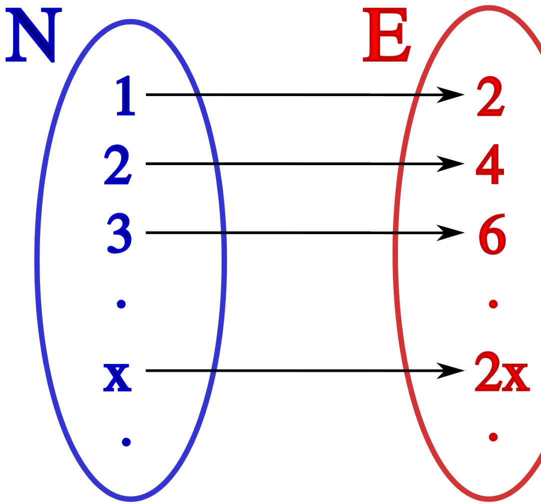
Also, since no even number is mapped twice, this function is bijective.



Cardinality of Odd and Even Numbers

Since bijection exists between ℕ and E (or O),

The cardinality of even numbers is same as cardinality of ℕ .



Countably Infinite Sets - Definition

- We say we can "**count**" the natural numbers because we can list them in order:

  $$\{\ 1,\ 2,\ 3,\ \ldots\}$$

- A set that has the **same cardinality** as $\mathbb{N}$ is called **countably infinite**.
- The cardinality of the natural numbers is often denoted by: $\aleph_0$

Questions

Which of the following sets have same cardinality as of N.

Extra: If they do, can you represent your bijection as an algebraic function?

1. $\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5} \ldots.\}$
2. The set of all square numbers less than a google.
3. Set of all integers

Cardinality of Integers

We can express the set of integers $\mathbb{Z}$ as the union of two sets:

- $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$ (positive natural numbers and 0)
- $\mathbb{Z}^- = \{-1, -2, -3, \ldots\}$ (negative integers)

To show that $\mathbb{Z}$ is countable, we define following **mapping**. This mapping pairs each natural number with a unique integer, proving that $\mathbb{Z}$ and $\mathbb{N}$ have the same cardinality.

n	0	1	2	3	4	5	6	7	8	9	10	$\cdots$
$f(n)$	0	1	-1	2	-2	3	-3	4	-4	5	-5	$\cdots$

Cardinality of ℕ × ℕ (Cartesian Product of ℕ)

		y					
		0	1	2	3	4	5
x	0	0	1	3	6	10	15
	1	2	4	7	11	16	22
	2	5	8	12	17	23	30
	3	9	13	18	24	31	39
	4	14	19	25	32	40	49
	5	20	26	33	41	50	60

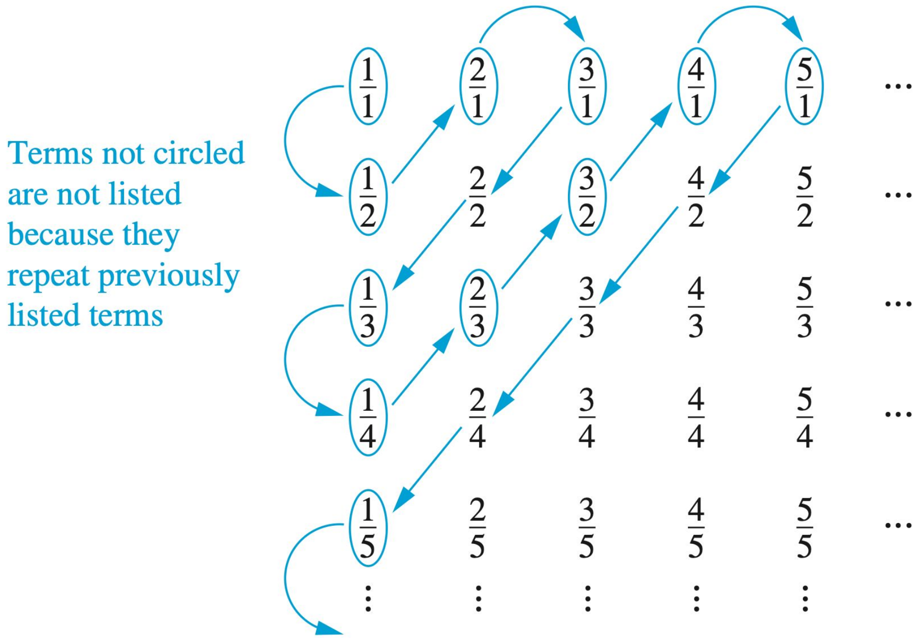
ℕ × ℕ denotes the cartesian product of natural numbers (ordered pair of natural number).

We show here a mapping from ℕ × ℕ to ℕ.

$$P(x, y) = \frac{(x+y)(x+y+1)}{2} + x,$$

This shows that the cardinality of ℕ × ℕ to ℕ is same.

Cardinality of Rational Numbers



We previously saw rationals as

$$\mathbb{Q} = \{\, a / b : \ a, b \in \mathbb{Z} \,\}$$

Here's a mapping of rational numbers to natural numbers.

We skip the repeated pairs here.

Decimal Strings

- In the **decimal system**, we use **10 symbols** (0–9) to represent numbers.
- In programming, there's a crucial difference:
  - The number `12` represents the **value** twelve — a count or quantity.
  - The string `"12"` represents a **sequence of characters**: the character `'1'` followed by the character `'2'`.
- Although they look similar, a **number** and its **string representation** are fundamentally different types of objects.

Ponder

So why does your calculator correctly compute:

$$12 + 34 = 46$$

…and not simply join them as `"1234"`?

Because there exists a **mapping** that translates strings like `"12"` into their numeric values so calculations can be performed correctly.

Decimal Strings and Natural Numbers

- Consider the set of all **strings** formed from the digits: {0,1,2,...., 9}
- With the constraint that the **first character is not 0**, each such string uniquely represents a **natural number** in the decimal system.
- For example:
  - `"7"` → 7
  - `"123"` → 123
  - `"42"` → 42

Decimal Strings and Natural Numbers

This means there's a **bijection** between:
Set of all finite decimal strings without leading zeros & ℕ

Therefore, the set of all such decimal strings has the **same cardinality** as the set of natural numbers.

Writing	"1"	"2"	"3"	…	"12"	…
ℕ	o	o o	o o o		o o o o o o o o o o o o	

Binary Strings

- If we change our number system from **decimal** to **binary**, all previous ideas still apply.
- A **binary string** is a sequence of characters from the set: **{0, 1}**
- Every finite binary string corresponds to a **natural number** when interpreted in the binary numeral system.
- For example:
  - `"101"` → 5
  - `"1101"` → 13

Ponder

- Every computer functions input and output are represented in binary string.
- Meaning every computer function’s input and output can be represented by corresponding natural numbers.

  ```
  const twice = (x) => {2*x}
  ```

- Does it means computers can calculate every well defined number ?

Enumeration - Basics

Let,

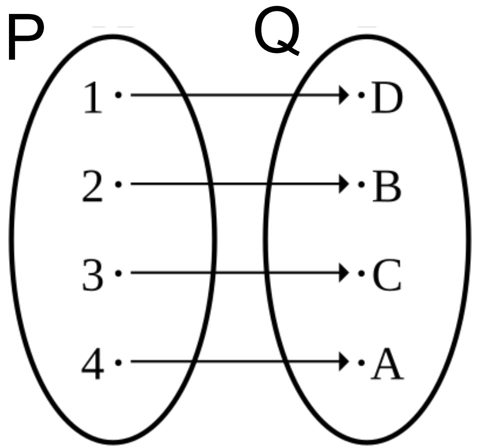
$Q = \{ a_1, a_2, a_3, \ldots.., a_n \}$ (a finite set)

$P = \{ 1, 2, 3, \ldots.., n \}$ (first n natural numbers)

Define $f : P \rightarrow A$,

$f(i) = a_i$

Then we have established a **bijection** between P and A.
This type of mapping is important because it **directly relates cardinality** to the number of elements in a finite set.



Enumeration

This process of mapping

$$M \rightarrow A$$

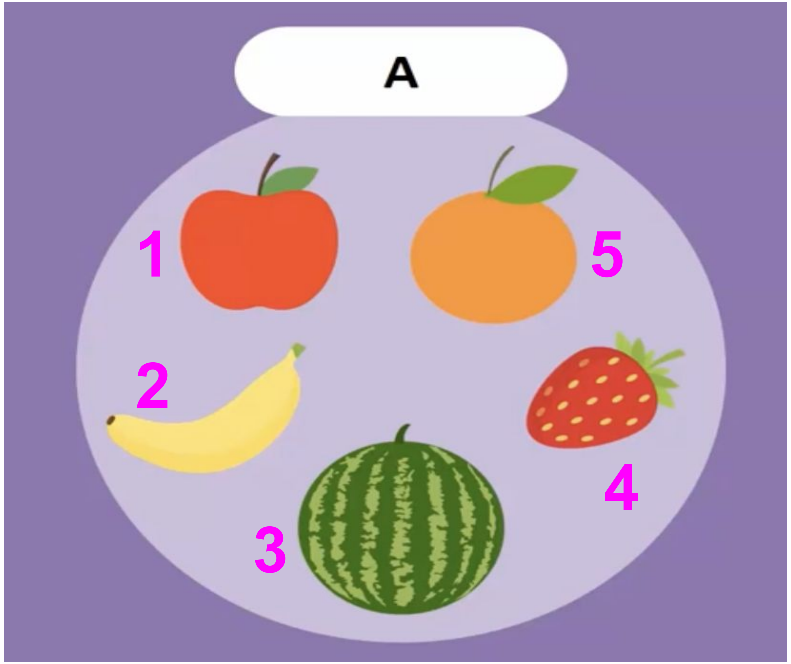
where

M = Set of first |A| natural numbers

A = Any Set

is called **enumeration**.

Think of it as numbering your sets.



Are all sets enumerable?

- Computer functions take some **input** and produce an **output**.
- Since every input can be converted into a **binary string**, every output produced by a pure computer function belongs to some **enumerable set**.
- This means that **computers can only generate sets that are enumerable**.
- Are there numbers or objects that a computer can't ever generate?

Sets that are not enumerable are called non enumerable sets.

Non Enumerable Set

To prove existence of non enumerable sets.

Mathematician Georg Cantor gave all set theorist a very powerful technique that has been used many time in history. It is called **Cantor's diagonal argument.**

This technique has been used various times to prove existence of something outside of given set. Let’s see it in action.

Listing all real numbers between 0 and 1

Cantor’s Diagonal Argument

Uncountably Infinite sets

- We have discovered that some sets have a **cardinality strictly larger** than that of the natural numbers.
- These are sets whose elements **cannot even be fully listed**—no matter how long we try.
- Such sets are described as **uncountable**.

A set with cardinality greater than that of $\mathbb{N}$ is called an **uncountably infinite set**.

Cardinality of Real Number

We've found that the **cardinality of the real numbers in (0, 1)** is greater than that of the rational numbers.

Consider $f(x) = 1/x$ for all $x \in (0,1)$

This function is *one-to-one*, and its range is $(1,\infty)$.
Therefore, the interval **(0,1)** has the same cardinality as **$(1,\infty)$**.

Without loss of generality we can say that,

Cardinality of $\mathbb{R}^+$ and $\mathbb{R}$ is same as that of interval (0,1).

Cardinality of Real Number

Cardinality of $\mathbb{R}$ is usually denoted by $\mathfrak{c}$. (*fraktur c*)

It can be shown that

$$\mathfrak{c} = 2^{\aleph_0}$$

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Quiz Quiz Quiz


Feedback Time

See You Guys in Next Session :)

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